Article

Vacuum birefringence and the polarized X-ray emission from a radio magnetar

https://doi.org/10.1038/s41586-026-10859-z

Received: 19 September 2025

Accepted: 29 June 2026

Published online: 5 August 2026

Check for updates

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Magnetars are isolated neutron stars with exceptionally strong surface fields exceeding 10^{14} G (ref. 1). Their bright X-ray emission probes physical regimes in which quantum electrodynamics (QED) influences radiation propagation[2–4]. Strong magnetic fields induce polarization-dependent refractive indices in the vacuum[5,6]; such vacuum birefringence remains a long-standing but unconfirmed prediction of QED. Here we report phase- and energy-resolved polarization measurements of the radio-emitting magnetar 1E 1547.0−5408 obtained by coordinating X-ray and radio observations from the Imaging X-ray Polarimetry Explorer, the Neutron Star Interior Composition Explorer and the Parkes/Murriyang Observatory. We detect large polarization degrees (PDs) in the thermally dominant soft X-ray band, reaching phase-averaged values of 65% at 2 keV before substantially decreasing between 2 keV and 4 keV. At certain rotational phases, the 2–3 keV PD rises to nearly 80% while remaining high (≳40%) throughout the radio beam crossing. The phase-dependent X-ray and radio polarization angles are both consistent with the rotating vector model, suggesting that the emission geometries track the large-scale magnetic field of the star. Collectively, these characteristics challenge standard surface emission models using non-refractive propagation of light to infinity. Vacuum-birefringence-governed magnetospheric propagation can naturally explain the X-ray polarization signals. Our results represent a marked advance in probing this hallmark prediction of QED, opening a new cosmic window into superstrong-field quantum physics, thereby motivating further observational and theoretical studies concentrating on this domain.

Magnetars[1,7,8], a class of isolated neutron stars (NSs), possess the strongest known magnetic fields in the observable universe ($B_{\rm surface}$ is of order $10^{14–15}$ G), surpassing the typical field strength of young, rotation-powered pulsars by one to three orders of magnitude. The decay of these magnetic fields powers persistent, high-luminosity X-ray emission, which typically peaks between 0.5 keV and 10 keV and reaches $L_X \gtrsim 10^{35}$ erg s^{-1} (see ref. 9 for a review). With magnetic field strengths that far exceed the quantum electrodynamic (QED) critical field $B_{\rm cr} = 4.4 \times 10^{13}$ G, magnetars are widely recognized as promising astrophysical laboratories for testing exotic, non-linear QED processes in the strong B-field regime[10,11].

An example of this prediction is vacuum birefringence (VB), in which intense magnetic fields—comparable to those of magnetars—substantially modify the refractive indices of photon polarization eigenmodes and render them disparate: different polarizations propagate at different speeds[12–14]. This property is a direct consequence of the polarization of the quantum vacuum, and its constituent virtual electron–positron pairs, by the electromagnetic field[5,6], and remains an unverified prediction of QED. Indirect evidence consistent with the action of VB has been reported both in laboratory contexts[15] and in astrophysical settings[16–18], whereas direct measurements in either of these settings have remained challenging[11,16,19–21]. Together, these complementary approaches provide important progress towards testing this fundamental physics prediction and offer promising avenues for further advances.

The influence of VB in magnetar magnetospheres is expected to strongly enhance the polarization signature of soft X-ray photons

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emerging from their surfaces. These photons are predicted to undergo differential phase shifts depending on their two polarization modes—ordinary (O mode) and extraordinary (X mode). If **k** represents the momentum vector of a photon, the O mode (∥) has an electric field vector approximately in the **k**–**B** plane, and the X mode (⊥) has its electric field perpendicular to this plane. VB decouples the polarization eigenmodes out to altitudes of about 30–300 stellar radii[22], leading to a tight correlation of their polarization vectors with the magnetospheric field morphology. The result is that VB enhances the net linear polarization of surface emission, potentially producing polarization degrees (PDs) approaching 100% under ideal viewing geometries[3,22,23]. Inside magnetar atmospheres, the combined influence of vacuum and plasma birefringence allows photons to convert between polarization modes[2,21]. This mode conversion is most efficient at a QED vacuum resonance (VR) energy (typically around a few keV) that depends on the hydrogenic plasma density, the local field strength and light propagation direction, and acts to depolarize the net PD emerging from the surface. Consequently, we expect a rotational-phase-averaged high PD away from the VR energy and pronounced PD reduction near it due to increased mode mixing.

1E 1547.0–5408 is a canonical magnetar with a spin period of 2.09 s and a surface equatorial magnetic field strength of about 2.2×10^{14} G (ref. 24). At present, it is in a persistently bright X-ray state, exhibiting a typical magnetar spectrum—a soft thermal-like component prominent in the $\lesssim$10 keV band, and a hard X-ray tail dominating at higher energies[25,26]. The broadband 0.5–70 keV spectrum is best fit with either a double blackbody (BB) thermal contribution and a non-thermal power-law (PL) model, or a single BB and double PL model[25,26]. The pulse shape at all energies is virtually sinusoidal, with minimal contribution from higher harmonics. The root-mean-squared pulsed fraction[27] is notably large compared with the rest of the magnetar population—it also exhibits a strong energy dependence with a value of approximately 30% at 2 keV that steadily increases to 42% around 6 keV, before decreasing monotonically towards 20% at energies >10 keV (refs. 25,26). These spectral and timing characteristics probably indicate the presence of a thermal hotspot on the surface dominating the softest energies of the spectrum up to around 6 keV, whereafter emission of magnetospheric origin overtakes the surface contribution of the spectrum and alters the pulse properties[25,26].

1E 1547.0–5408 is also unique among the magnetar population as a persistent radio pulsar[28]. It emits broad radio pulses at the spin period of the magnetar, spanning approximately 25% of the source duty cycle (Methods), unusually wide among the young, isolated NS population. The pulses are also highly polarized and exhibit a characteristic S-shaped swing in the linear polarization position angle[28], indicative of the viewing line of sight crossing the radio emission cone[29]. These properties hint at a nearly aligned rotator, with an observer line of sight at a small angle to the rotation axis that samples open magnetic field lines proximate to the polar cap (Methods).

This richness of information offered by the combined X-ray + radio data motivated a deep simultaneous observing campaign with the Imaging X-ray Polarimetry Explorer (IXPE), the Neutron star Interior Composition Explorer (NICER) and Murriyang, the 64-m Parkes radio telescope, which was carried out between 26 March 2025 UTC 03:01:52 and 5 April 2025 UTC 00:26:53. Our ultimate goal is to study in detail the soft (2–8 keV) X-ray polarimetric properties of the source as afforded by IXPE and to place it in context of the geometrical constraints achievable through the radio observations. Hence, all of the observations were coordinated to optimize simultaneous coverage between the instruments, particularly with IXPE (see the Methods and Supplementary Information for full details of the observations).

The IXPE data show an energy- and phase-averaged 2–8 keV PD of about 46 ± 4% (1σ uncertainty; here and throughout the rest of the text) with a polarization angle (PA) of −76 ± 3° (Fig. 1) with a statistical significance of about 10.5σ above the minimum detectable polarization

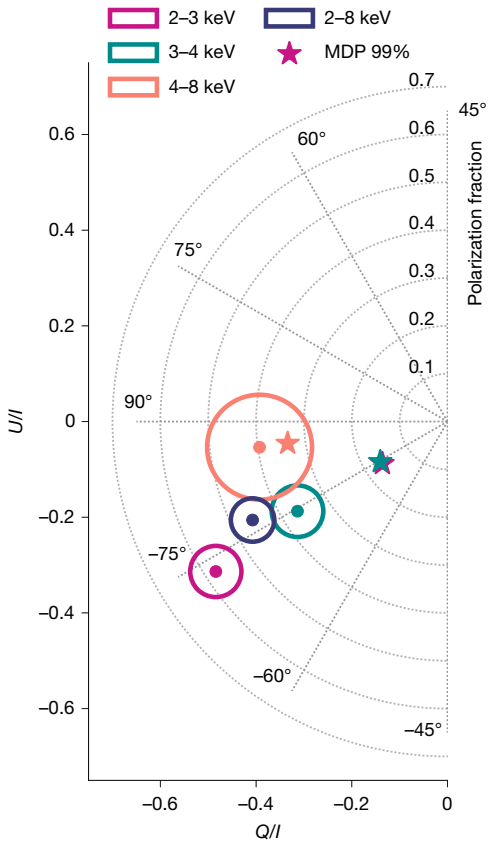


Fig. 1 | Energy-resolved IXPE polarization measurements. Background-subtracted and phase-averaged polarization measurements of 1E 1547.0–5408. We show the PD in four energy bands, 2–3 keV, 3–4 keV, 4–8 keV and 2–8 keV. Contours show the 68.3% (1σ) confidence regions for Stokes Q/I and U/I (PD and PA are shown in the radial and polar lines, respectively). The stars indicate the corresponding minimum detectable polarization at the 99% confidence level (MDP_{99}) for each energy bin. The 2–8 keV energy-integrated polarization characteristics are shown in dark blue. Notably, the energy-integrated PD (dark blue) is 46% and the 2–3 keV PD (magenta) reaches 59%. The PD decreases substantially in the 3–4 keV energy range. The PD remains high at 40% in the 4–8 keV band, yet it suffers from a considerably lower S/N. The PA does not exhibit strong variation across energy.

value for the 99% confidence level (MDP_{99}). This is one of the largest PDs in the population of six IXPE-observed magnetars[11,30,31]. The PD also exhibits strong energy dependence, achieving a maximum at 59 ± 5% in the softest energy band of 2–3 keV before decreasing to 37 ± 5% in the 3–4 keV range. By contrast, the PA of the two bands remain consistent with that of the energy-integrated measurement. In the hard part of the energy range (4–8 keV), we measure a PD of 40 ± 11%, and slightly shifted PA of −86 ± 8°, a detection at only the 3.5σ significance level, mainly because of its lower signal-to-noise ratio (S/N). Notably, the PD of the softest energy range is at least a factor of 2.5 larger than the corresponding values measured in the rest of the population[31].

Using a phase-coherent timing ephemeris that covers our full observing campaign, we phase-fold the X-ray (intensity and polarization) and radio data, after correcting for propagation through the ionized interstellar medium to ensure absolute alignment between the two wavelengths (Methods). The PD is most prominent in the softer part of the spectrum (Fig. 2). It modulates moderately with phase, reaching a peak of 74 ± 11% in the 2–4 keV range and a minimum of about 34 ± 8%; the former occurs close to the X-ray pulse minimum and away from the radio pulse, in contrast to the latter. Note that the PD reaches 82 ± 15% in the 2–3 keV range and a minimum of 42 ± 12% (Extended Data Fig. 6).

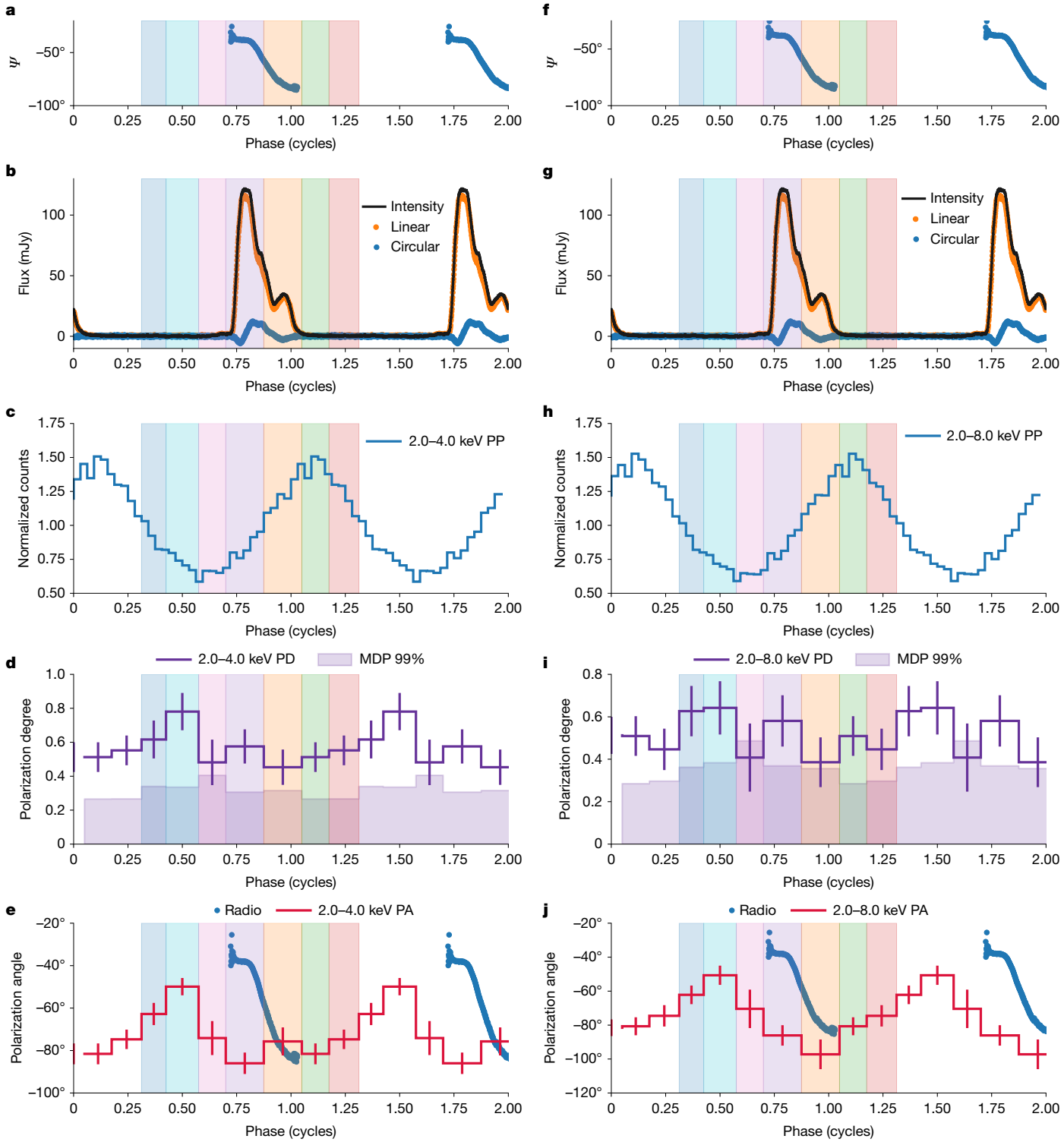


Fig. 2 | Phase-resolved radio and X-ray polarization characteristics of 1E 1547.0−5408. a,**f**, radio PA (two cycles shown for clarity). **b**,**g**, The radio intensity curve (black) along with the total linear radio polarization (orange) as well as the circular radio polarization (blue). All radio data shown are averaged across our three Murriyang observations. **c**–**e**,**h**–**j**, The IXPE intensity (**c**,**h**), PD (**d**,**i**) and PA (**e**,**j**) pulse profiles in the 2–4 keV and 2–8 keV bands, respectively. The radio PA is also shown in **e** and **j** for ease of comparison to the X-ray PA. The shaded curves overlaying the PD panels show the MDP_{99} at each corresponding phase bin. The vertical coloured bands highlight the binning used for the IXPE phase-resolved polarization analysis. All error bars correspond to 68.3% (1σ) uncertainties.

The PA displays a sinusoidal modulation between a maximum of −51 ± 6° and a minimum of −97 ± 9°. The overall modulation is comparable in amplitude and shape to that observed in the radio band, albeit leading by a phase offset of around 0.3. Notably, a similar phase offset is observed in the intensities, but now with the X-ray peak trailing the radio one.

A spectro-polarimetric analysis of the IXPE + NICER phase-averaged spectrum shows that the emission in the soft 1–8 keV X-ray band is best described by a BB component with a temperature $kT = 0.72 \pm 0.01$ keV and a relatively small emitting area $R^2 = 1.07 \pm 0.05^2$ km (see Table 1 and Methods), assuming a source distance of 4.5 kpc (ref. 32).

Table 1 | Best-fit spectro-polarimetric model parameters

Model parameter	Value ±1σ
N_H (cm^{-2})	3.67×10^{22} (fixed)
PD at 1 keV	0.80 ± 0.08
PD_{slope}	0.15 ± 0.02
PA at 1 keV (°)	-67^{+4}_{-7}
PA_{slope} (° keV^{-1})	−4 ± 4
BB kT (keV)	0.720 ± 0.005
R^2 (km^2)	1.07 ± 0.05
$C_{NICER\,Obs1}$	1.0 (fixed)
$C_{NICER\,Obs2}$	1.08 ± 0.05
$C_{NICER\,Obs3}$	1.05 ± 0.05
$C_{NICER\,Obs4}$	1.01 ± 0.05
C_{DU1}	0.80 ± 0.03
C_{DU2}	0.80 ± 0.03
C_{DU3}	0.76 ± 0.03
Total fit statistic	1,798.80 with 1,511 dof

Spectral and polarimetric parameters for the best-fit model, a single absorbed (column density, N_H) blackbody (BB) with linearly varying polarization degree (PD) and polarization angle (PA). A multiplicative cross-calibration uncertainty (labelled as $C_{instrument}$) is also added to the model to track instrumental calibration differences across the four NICER observations and the three IXPE detector units (DUs). All uncertainties are calculated at the 68.3% confidence level (1σ). The fit statistic uses a combination of the C-stat and χ^2 statistics used for NICER and IXPE, respectively. Blackbody emitting radius (R^2) is derived adopting a distance of 4.5 kpc (ref. 32). dof, degrees of freedom.

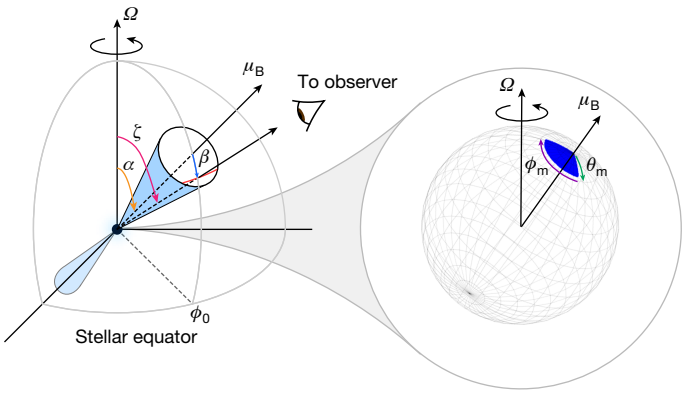


Fig. 3 | Illustration of a rotating NS with a dipole magnetic field geometry described by the RVM. The magnetic axis μ_B inclined from spin axis Ω by an angle α and an observer viewing angle from the spin axis ζ. The magnetic meridian is defined at a reference phase ϕ_0 (on left), for which the spin and magnetic axes and the observer line of sight are coplanar. The emission radio cone (medium blue) is centred around μ_B and its intersection with the observer line of sight is indicated by the red line. Our best-fit wedge-shaped hotspot corresponding to the light curves in Extended Data Fig. 8 is shown in blue in the inset on the right. The hotspot extent on the NS surface in the inset is constrained by defined magnetic co-latitudes $0 \le \theta_m \le 17°$ and longitudes ϕ_m ranging from the meridian to 120°.

In accentuating the model-independent polarization results presented above, we find that the phase-averaged BB polarization Q and U spectra are best modelled by a linearly decreasing PD from 65 ± 8% at 2 keV at a rate of −15 ± 2% keV^{-1}, and a constant PA of $-67°^{+4}_{-7}$. We do not find evidence for any trend of changing PA or PD at the upper end of the IXPE energy band (≳4 keV), as hinted at (albeit at a low significance) in our model-independent analysis. This is probably because the 2–8 keV IXPE source spectrum is heavily dominated by the 2–4 keV range, which contributes the most photons; adding a second component to the spectrum is not statistically justified and, when included, remains unconstrained (Methods).

The modest phase-averaged PD measured for most magnetars in the soft, 2–3 keV X-ray band (≲20%; ref. 31) has so far been interpreted as evidence for the presence of a condensed surface[18,33]. The much higher PD we measure in 1E 1547.0–5408 challenges such a scenario for this source. Given the high surface temperature reported here and elsewhere[25,26] and a dipole field strength of about 2×10^{14} G (ref. 24), magnetic condensation is unlikely for 1E 1547.0–5408 (ref. 34). Phase-averaged PDs approaching about 40% may be attainable in a pure magnetized atmosphere without invoking additional effects, but reproducing values of around 65% is non-trivial[4]. More challenging for such models are the extreme phase-resolved PDs (approximately 70–80% in the 2–4 keV band; Fig. 2) and the absence of strong depolarization near the radio pulse phase. Although we detect a minimum X-ray PD during the radio pulse, it remains comparatively high at about 30–40%, whereas substantial depolarization would be expected when sampling open field lines[35]. Together, these properties highlight tensions with standard surface and atmospheric models and suggest that additional physical processes influencing the propagation of polarized radiation through the magnetosphere may be at play.

The observed S-shaped sweep of the radio PA across the time and frequency averaged 2.5–4.0 GHz profile (Fig. 2, top) is consistent with the rotating vector model (RVM) for a dipolar magnetic field geometry[36] and is well fit by this model, yielding a nearly aligned rotator viewed close to the spin axis (Extended Data Fig. 5); see Fig. 3 for a schematic of the geometry. We recover a magnetic and spin axes that are offset by $\alpha = 3.4°^{+1.3}_{-1.2}$, along with the angle between the spin axis and our line of sight of $\zeta = 7.5°^{+3.0}_{-2.6}$. Under the simplifying assumption that the open field lines above the polar cap are fully illuminated, this combination of the magnetic and viewing geometry along with the high pulse duty cycle suggests the radio emission originates around 692^{+654}_{-400} km above the NS surface (Methods). This height is less than 1% of the light cylinder, indicating that the absolute radio pulse phase accurately describes a magnetic pole crossing.

The X-ray PA evolution can likewise be adequately described within the RVM formalism (Methods and Extended Data Fig. 7), indicating that the X-ray polarization follows a geometrical modulation consistent with the large-scale magnetic field structure. Although the constraints are weaker than in the radio because of the lower S/N, the inferred geometric parameters are broadly consistent with those obtained from the radio analysis at the approximately 2.5σ level, while not uniquely constraining the geometry[37]. In particular, the allowed parameter space remains large, with only a limited region overlapping the radio-derived constraints[37]. Differences between the radio and X-ray solutions, including phase offsets in the intensity and polarization profiles, suggest that the signals in the two bands originate from different locations relative to the magnetic axis of the star, and may arise from distinct emission altitudes or from deviations from a purely dipolar field configuration (Methods).

The decrease in PD within the narrow energy range of 2–4 keV is notable. A separate and differently polarized component contributing to the BB emission is not evident in the spectral and timing properties of the source, which argue for the same thermal component dominating these energies. This behaviour is most naturally attributed to mode conversion close to VR[2,21]. Given the B-field strength of 1E 1547.0–5408, and assuming a hydrogen atmosphere, mode conversion becomes efficient at energies ≳2 keV. At energies ≳4 keV, the interpretation becomes less robust because of the reduced S/N. Nevertheless, the 4–8 keV (and 4–5 keV; Methods) band shows a tentative deviation from the trend observed at lower energies. Deeper IXPE observations are essential to clarify the polarimetric behaviour in this regime.

The exceptionally high PDs reaching 65 ± 8% at 2 keV and rising to 82 ± 15% at certain rotational phases, the absence of strong depolarization near the radio pulse phase, the observed decrease in polarization

in the 2–4 keV range, and the coherent, RVM-like evolution of the PA across both X-ray and radio bands present a coherent observational picture. These elements, taken together, cannot be easily reconciled with the broad sampling of magnetic field directions represented in standard atmospheric models for extended surface X-ray hotspot locales. The signal properties are naturally interpreted in scenarios in which the propagation of polarized radiation is influenced by QED effects operating in the magnetosphere. In particular, the combination of high polarization fractions and modest PA modulation suggests that the polarization direction remains closely coupled to the large-scale magnetic field geometry over extended distances. Within this context, magnetospheric VB provides a natural physical mechanism to account for the observed polarization behaviour.

To illustrate the VB interpretation, we performed simulations of the X-ray intensity and polarization properties using the MAGTHOM-SCATT radiative transfer code[38–40] (Methods). Within this adopted modelling framework, configurations that include magnetospheric VB provide a markedly improved description of the phase-resolved *I*, *Q*, *U* polarization data compared with models that neglect it (Methods and Extended Data Fig. 8): non-refractive propagation yields notably stronger phase-dependent variations in Stokes *Q* and *U* than are present in the data. Constrained by pulse phase offsets between the radio and X-ray signals (Fig. 2), the best-fit model indicates that the centroid of the X-ray hotspot is offset from the magnetic pole (Fig. 3). Although these results are model-dependent, they provide additional support for the interpretation that magnetospheric propagation effects play a key part in shaping the observed polarization signatures in 1E 1547.0–5408.

The combined X-ray and radio observational results presented here highlight the growing ability to probe QED effects in the extreme magnetic field environments of magnetars. Future X-ray polarimetric observations, particularly at lower energies at which even higher PDs may be expected, are essential for further advancing this effort. Continued observations by IXPE, prospects afforded by future soft X-ray polarimetric missions such as GoSOX[41], together with laboratory and collider-based experiments, will enable increasingly stringent tests of the most fundamental predictions of QED.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-10859-z.

1. Kouveliotou, C. et al. An X-ray pulsar with a superstrong magnetic field in the soft γ-ray repeater SGR1806 - 20. *Nature* **393**, 235–237 (1998).
2. Lai, D. & Ho, W. C. G. Polarized X-ray emission from magnetized neutron stars: signature of strong-field vacuum polarization. *Phys. Rev. Lett.* **91**, 071101 (2003).
3. van Adelsberg, M. & Perna, R. Soft X-ray polarization in thermal magnetar emission. *Mon. Not. R. Astron. Soc.* **399**, 1523–1533 (2009).
4. Taverna, R., Turolla, R., Suleimanov, V., Potekhin, A. Y. & Zane, S. X-ray spectra and polarization from magnetar candidates. *Mon. Not. R. Astron. Soc.* **492**, 5057–5074 (2020).
5. Heisenberg, W. & Euler, H. Folgerungen aus der Diracschen Theorie des Positrons. *Z. Phys.* **98**, 714–732 (1936).
6. Schwinger, J. On gauge invariance and vacuum polarization. *Phys. Rev.* **82**, 664–679 (1951).
7. Duncan, R. C. & Thompson, C. Formation of very strongly magnetized neutron stars - Implications for gamma-ray bursts. *Astrophys. J. Lett.* **392**, L9–L13 (1992).
8. Paczynski, B. GB 790305 as a very strongly magnetized neutron star. *Acta Astronaut.* **42**, 145–153 (1992).
9. Kaspi, V. M. & Beloborodov, A. M. Magnetars. *Annu. Rev. Astron. Astrophys.* **55**, 261–301 (2017).
10. Harding, A. K. & Lai, D. Physics of strongly magnetized neutron stars. *Rep. Prog. Phys.* **69**, 2631–2708 (2006).
11. Taverna, R. & Turolla, R. X-ray polarization from magnetar sources. *Galaxies* **12**, 6 (2024).
12. Adler, S. L. Photon splitting and photon dispersion in a strong magnetic field. *Ann. Phys.* **67**, 599–647 (1971).
13. Tsai, W.-Y. & Erber, T. Propagation of photons in homogeneous magnetic fields: index of refraction. *Phys. Rev. D* **12**, 1132–1137 (1975).
14. Pavlov, G. G. & Gnedin, Y. N. Vacuum polarization by a magnetic field and its astrophysical manifestations. *Astrophys. Space Phys. Res.* **3**, 197 (1984).
15. Adam, J. et al. Measurement of e^+e^- momentum and angular distributions from linearly polarized photon collisions. *Phys. Rev. Lett.* **127**, 052302 (2021).
16. Ho, W. C. G., Lai, D., Potekhin, A. Y. & Chabrier, G. Atmospheres and spectra of strongly magnetized neutron stars. III. Partially ionized hydrogen models. *Astrophys. J.* **599**, 1293–1301 (2003).
17. Mignani, R. P. et al. Evidence for vacuum birefringence from the first optical-polarimetry measurement of the isolated neutron star RX J1856.5-3754. *Mon. Not. R. Astron. Soc.* **465**, 492–500 (2017).
18. Taverna, R. et al. Polarized X-rays from a magnetar. *Science* **378**, 646–650 (2022).
19. Ho, W. C. G. & Lai, D. Atmospheres and spectra of strongly magnetized neutron stars - II. The effect of vacuum polarization. *Mon. Not. R. Astron. Soc.* **338**, 233–252 (2003).
20. Ejlli, A. et al. The PVLAS experiment: a 25 year effort to measure vacuum magnetic birefringence. *Phys. Rep.* **871**, 1–74 (2020).
21. Lai, D. IXPE detection of polarized X-rays from magnetars and photon mode conversion at QED vacuum resonance. *Proc. Natl Acad. Sci. USA* **120**, e2216534120 (2023).
22. Heyl, J. S., Shaviv, N. J. & Lloyd, D. The high-energy polarization-limiting radius of neutron star magnetospheres - I. Slowly rotating neutron stars. *Mon. Not. R. Astron. Soc.* **342**, 134–144 (2003).
23. Heyl, J. S. & Shaviv, N. J. Polarization evolution in strong magnetic fields. *Mon. Not. R. Astron. Soc.* **311**, 555–564 (2000).
24. Camilo, F., Ransom, S. M., Halpern, J. P. & Reynolds, J. 1E 1547.0–5408: a radio-emitting magnetar with a rotation period of 2 seconds. *Astrophys. J.* **666**, L93–L96 (2007).
25. Coti Zelati, F. et al. The long-term enhanced brightness of the magnetar 1E 1547.0–5408. *Astron. Astrophys.* **633**, A31 (2020).
26. Lower, M. E. et al. The 2022 high-energy outburst and radio disappearing act of the magnetar 1E 1547.0–5408. *Astrophys. J.* **945**, 153 (2023).
27. Bildsten, L. et al. Observations of accreting pulsars. *Astrophys. J. Suppl. Ser.* **113**, 367–408 (1997).
28. Camilo, F., Reynolds, J., Johnston, S., Halpern, J. P. & Ransom, S. M. The magnetar 1E 1547.0–5408: radio spectrum, polarimetry, and timing. *Astrophys. J.* **679**, 681–686 (2008).
29. Zeng, S., Philippov, A., Juno, J., Beloborodov, A. M. & Popova, E. Origin of pulsed radio emission from magnetars. *Astrophys. J. Lett.* **996**, L20 (2026).
30. Stewart, R. et al. X-ray polarization of the magnetar 1E 1841-045. *Astrophys. J. Lett.* **985**, L35 (2025).
31. Rigoselli, M. et al. IXPE detection of highly polarized X-rays from the magnetar 1E 1841-045. *Astrophys. J. Lett.* **985**, L34 (2025).
32. Tiengo, A. et al. The dust-scattering X-ray Rings of the anomalous X-ray pulsar 1E 1547.0–5408. *Astrophys. J. Lett.* **710**, 227–235 (2010).
33. Heyl, J. et al. The detection of polarized X-ray emission from the magnetar 1E 2259+586. *Mon. Not. R. Astron. Soc.* **527**, 12219–12231 (2024).
34. Medin, Z. & Lai, D. Condensed surfaces of magnetic neutron stars, thermal surface emission, and particle acceleration above pulsar polar caps. *Mon. Not. R. Astron. Soc.* **382**, 1833–1852 (2007).
35. Dinh Thi, H., Baring, M. G., Hu, K., Harding, A. K. & Barchas, J. A. Monte Carlo simulations of polarized radiative transfer in neutron star atmospheres. *Astrophys. J.* **992**, 188 (2025).
36. Radhakrishnan, V. & Cooke, D. J. Magnetic poles and the polarization structure of pulsar radiation. *Astrophys. Lett.* **3**, 225 (1969).
37. Li, B.-P., Gao, Z.-F., Ma, W.-Q. & Zhang, W.-F. Probing the origin of magnetar X-ray polarization diversity: a multi-wavelength geometrical study of 1E 1547.0-5408 and 1E 2259+586. Preprint at arxiv.org/abs/2604.10477 (2026).
38. Barchas, J. A., Hu, K. & Baring, M. G. Polarized Radiation Transfer in Neutron Star Surface Layers. *Mon. Not. R. Astron. Soc.* **500**, 5369–5392 (2021).
39. Hu, K., Baring, M. G., Barchas, J. A. & Younes, G. Intensity and polarization characteristics of extended neutron star surface regions. *Astrophys. J.* **928**, 82 (2022).
40. Dinh Thi, H. et al. Pulsed, polarized X-ray emission from neutron star surfaces: the effects of vacuum birefringence in the magnetosphere. *Astrophys. J.* **1000**, 73 (2026).
41. Marshall, H. L. et al. The globe orbiting soft X-ray (GOSoX) polarimeter concept study. In *Proc. SPIE 11822, Optics for EUV, X-Ray, and Gamma-Ray Astronomy X* Vol. 11822 (SPIE, 2021).

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Methods

X-ray spectro-polarimetric analysis

We performed spectro-polarimetric analysis of the 0.5–12 keV NICER and 2–8 keV IXPE spectral data using Xspec v.12.15.0 to investigate the nature of the source emission. For the NICER spectra, we grouped each spectral bin with a minimum of 5 counts per bin and for IXPE spectra, with a minimum of 30 counts per bin and accordingly performed statistical fits with the *W*-statistic (χ^2) using C-stat (χ) within Xspec. Although the energy channels typically contain substantially more counts than these minimum thresholds, the adopted grouping ensures that the lowest and highest energy bins remain in the appropriate statistical regime for each instrument. Given the low S/N of the source at energies $\lesssim$2 keV, we froze the column density ($N_{\rm H}$) at 3.67×10^{22} cm^{-2}, as reported in ref. 26; we adopted the Wilm solar abundances[42] and Verner cross-sections[43] to preserve consistency with the previous analysis. Finally, our spectral modelling imposes a constant between each telescope to track cross-instrument calibration uncertainties (Table 1).

We simultaneously fit the NICER SCORPEON background model with the source spectra. While fitting, we fix the trapped-electron component normalizations to 0 because they negligibly contribute to the fitted background. The background models show that the NICER source spectra are background-dominated below about 1.4 keV and above approximately 7 keV. Likewise, the background dominates the IXPE spectra above about 6 keV, limiting the statistics of the higher energy contributions.

Historically, the broadband spectrum has been well described using either a 2BB + PL or a BB + 2PL fit[25,26]. However, the limited statistics of the source data constrain the detectability of the hot BB or PL components, which emerge at ≥6 keV. We performed an *F*-test for a single and double BB model and found an *F*-statistic value of 2.47 and an associated probability of 0.09. Whereas the *F*-test favours including the second BB component, the 2BB best-fit model yields physically unrealistic values. We thus reject the extra component in favour of a single absorbed BB model for this particular dataset, with a total fit statistic of 1,798.80/1,511 degrees of freedom (dof) (Extended Data Figs. 2 and 3). Moreover, we favour applying a linear polarization component to the model with a total fit statistic of 1,798.80/1,511 dof compared with 1,946.38 for a PL component and 1,840.09 for a constant component for 1,513 dof. A single absorbed PL model yielded largely unphysical parameters and thus was rejected.

A significant detection was found in the 4–8 keV band of the model-independent analysis (Fig. 1). However, the high-energy component possesses large uncertainties, impeding comparisons to the 2–3 keV and 3–4 keV behaviour and in discerning trends in the energy-resolved polarization characteristics. We further investigated the high-energy component by subdividing the energy bands to 4–5 keV and 5–8 keV, similar to the analysis presented in ref. 44. We analysed these bins comparatively using a model-independent approach in ixpeobssim and a model-dependent approach in Xspec, in which each energy bin was fit according to an absorbed BB model with a constant polarization component. We fixed the BB model parameters to their best-fit value obtained from the energy-integrated NICER + IXPE spectral fits. This bin-by-bin spectro-polarimetric approach is standard practice in IXPE analyses and effectively yields model-independent polarization measurements. The Xspec error contours were obtained by Monte Carlo sampling of the best-fit parameter confidence levels and transforming the resulting distributions into Stokes *Q*–*U* space. The ixpeobssim and Xspec analyses yield a PD of approximately 50 ± 10% and 48 ± 15%, respectively, and the ixpeobssim analysis reveals that the 4–5 keV band exhibits a statistically significant detection at the approximately 5.5σ level (Extended Data Fig. 1). Although the 4–5 keV values appear to deviate from the lower-energy trend, their large uncertainties remain consistent with the 2–3 keV and 3–4 keV bands. The 2–4 keV band dominates the observed counts, whereas the 4–8 keV band contributes about 10%, possibly explaining why the linear polarization model fits the 2–8 keV data well. A deeper exposure is required to constrain the data above 5 keV, where there is likely a contribution from another spectral component[25,26].

We use the Goodman–Weare Markov chain Monte Carlo method within Xspec with 10 walkers drawn from a multidimensional Gaussian centred on the best-fit parameters. Three chains of 50,000 simulated spectral fits were generated (totalling 150,000) with the first 5,000 steps from each chain burned. The resulting posterior distributions were used to generate one-dimensional and two-dimensional corner plots and assess parameter correlations, shown in Extended Data Fig. 4. We used the standard uniform priors of Xspec for each model parameter, which sufficiently sampled the full physically motivated parameter space apart from the PA slope, in which we expanded the previous limits from ±5° keV^{-1} to ±30° keV^{-1}.

Radio polarization analysis

Assuming the radio emission from 1E 1547.0–5408 originates from above the polar cap of a dipole magnetic field component[29], we can use the RVM to constrain the magnetic and viewing geometry of the magnetar[36]. The linear PA sweep can be described as

$$\tan(\psi - \psi_0) = \frac{\sin\alpha\sin(\phi - \phi_0)}{\sin\zeta\cos\alpha - \cos\zeta\sin\alpha\cos(\phi - \phi_0)}, \quad (1)$$

where ψ_0 is the PA at the inflection point of the RVM swing, ϕ is the rotation phase of the magnetar, ϕ_0 is the phase at which the PA inflection occurs, α is the magnetic inclination angle and $\zeta = \alpha + \beta$ is the viewing angle between the spin axis and our line of sight, which is the product of the magnetic inclination angle and impact parameter (β). Note that the PA swing of the RVM is inverted, that is, $\psi_{\rm obs} = -\psi_{\rm RVM}$ in the observer reference frame[45]. We fit the RVM to the average profile obtained from combining our three concurrent Murriyang observations (see Extended Data Tables 1 and 2 for observation timings and IDs) using the bilby Bayesian inference library[46] as an interface to the dynesty nested sampling algorithm[47], and a Gaussian likelihood of the form

$$\mathcal{L}(d|\theta) = \prod_i^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left[-\frac{(d_i - \mu(\theta))^2}{2\sigma_i^2}\right], \quad (2)$$

where d is the data (corresponding to the PA), $\mu(\theta)$ is the RVM with model parameters given by θ, and σ is the uncertainty on the PA. Note, we modified the formal uncertainties on the PA by adding in an additional error in quadrature term as $\sigma^2 = \sigma_\psi^2 + \sigma_{\rm EQUAD}$ to account for potential underestimation of the PA uncertainties due to uncorrected systematic effects. We assumed priors on α and ζ that were uniform in solid angle, that is, where ($\pi(\theta) \propto \sin\theta$). Standard uniform priors were used for ϕ_0 and ψ_0. Our prior bounds for each parameter were set between $0° < \pi(\alpha) < 180°$, $0° < \pi(\zeta) < 180°$, $270° < \pi(\phi_0) < 360°$ and $-90° < \pi(\psi_0) < 0°$. Note that the restricted prior range on ϕ_0 is required to break a symmetry in the RVM that results from the unknown rotational direction (prograde or retrograde) of the magnetar. The resulting one-dimensional and two-dimensional posterior distributions are shown in Extended Data Fig. 5.

Our recovered values of $\alpha = 3.4^{\circ}{}^{+1.3}_{-1.2}$ and $\zeta = 7.5^{\circ}{}^{+3.0}_{-2.6}$ (equating to $\beta = 4.2^{\circ}{}^{+1.7}_{-1.5}$) indicate 1E 1547.0–5408 is an aligned rotator, viewed almost pole-on. This geometry is seemingly in tension with the previously reported values of $\alpha = 160°$ and $\beta = 14°$ (ref. 28). However, it can be largely explained by the covariance between the magnetic and viewing angles because of the unknown spin direction of the magnetar, and a potential usage of the incorrect PA sign convention in their RVM fit. Re-fitting the PA swing of the 2007 observation analysed in ref. 28, we obtained values of $\alpha = 10.2^{\circ}{}^{+3.9}_{-3.5}$ and $\zeta = 17.7^{\circ}{}^{+6.7}_{-6.1}$, for a magnetic impact angle of $\beta = 7.5^{\circ}{}^{+2.9}_{-2.6}$. This updated fit is consistent with our 2025 measurement to within the 68–95% confidence intervals. When the incorrect PA sign

convention is used, we obtain similarly large values of $\alpha = 169.7^{\circ}{}^{+3.5}_{-4.0}$ and $\zeta = 162.2^{\circ}{}^{+6.1}_{-2.6}$ to those of ref. 28.

An estimate of the radio emission height can be inferred from a combination of the profile phase width (W) and the measured magnetic and viewing geometry. Assuming a filled emission cone, the beam-opening angle (ρ) can be computed as[48]

$$\cos\rho = \cos\alpha\cos\zeta + \sin\alpha\sin\zeta\cos(W/2). \quad (3)$$

The opening angle is related to the emission height ($h_{\rm em}$) using[49]

$$\rho = 3\sqrt{\frac{\pi\, h_{\rm em}}{2Pc}}, \quad (4)$$

where P is the magnetar rotation period and c is the vacuum speed of light. Typically, the value of W is taken to be either the profile width at the 10% or 50% level (that is, W_{10} or W_{50}). However, given the breadth of the pulse peak of 1E 1547.0−5408 (Fig. 2), after iterating over different on-pulse window sizes, we settled on a 1.5% profile width $W = W_{1.5} = 142^{\circ}$. Combining this value with our measurements of α and ζ, we obtain $h_{\rm em} = 692^{+654}_{-400}$ km. This is broadly consistent with emission heights inferred for rotation-powered pulsars[50], yet substantially lower than those deduced for other magnetars[51,52].

X-ray PA modelling

We model the X-ray PA swing using the RVM following the same formalism adopted for the radio analysis (equation (1)). The data are restricted to the 2–4 keV energy band, in which the S/N is the highest (Fig. 1).

We performed a Bayesian inference of the RVM parameters with the ensemble sampler emcee[53]. Following previous IXPE analyses of X-ray pulsars[54], we used the wrapped-angle likelihood of ref. 55. This likelihood is specifically designed to handle the non-Gaussian uncertainties inherent to PA measurements and remains statistically valid in the presence of low-S/N data points. The phase-resolved PA profile is binned into 15 rotational bins. We used the same priors as the radio RVM fits, albeit with extended prior ranges on ψ_0 and ϕ_0 of $-180^{\circ} < \pi(\psi_0) < 0^{\circ}$ and $0^{\circ} < \pi(\phi_0) < 360^{\circ}$.

The RVM provides an adequate description of the data, with $\chi^2 = 12.8$ for 11 dof ($\chi^2_\nu = 1.16$). The resulting posterior distributions are shown in Extended Data Fig. 7. Given the lower photon statistics compared with the radio data, the X-ray constraints are broader, particularly for the viewing angle $\zeta = 69^{\circ}{}^{+25}_{-23}$ (Extended Data Fig. 7). The inferred geometric parameters are broadly consistent with those obtained from the radio RVM fit at the approximately 2.5σ level. However, the X-ray data do not uniquely constrain the geometry. In particular, the allowed parameter space remains large, with only a limited region overlapping the radio-derived constraints. Within these uncertainties, we recover a small magnetic inclination angle, $\alpha = 18 \pm 4^{\circ}$ (68% confidence interval), consistent with a quasi-aligned geometry. To assess the robustness of this result, we repeated the analysis using phase bins ranging from 8 to 16, obtaining consistent posterior distributions in all cases. We further validated our findings using an unbinned likelihood formalism[56], which yielded fully consistent parameter estimates. We note that recent Bayesian RVM modelling of the X-ray PA swing in 1E 1547.0−5408, using priors that are informed by the radio best-fit parameters, returns posteriors that are fully compatible with the radio-inferred geometry, that of a quasi-aligned rotator[37].

We find a large and statistically significant difference between the X-ray and radio fit for ϕ_0 amounting to $\Delta\phi_0 \approx 80^{\circ}$. This may be attributed to several contributing factors. For uniform cone emission, the radio ϕ_0 marks the rotational phase at which the magnetic axis is closest to the line of sight, which arises at the meridional plane where the spin axis, magnetic axis and line of sight of the observer are coplanar. By contrast, the X-ray surface spot for the wedge case, for example, need not be uniform in magnetic longitude and with a centroid that is offset from the pole. This angular displacement between the X-ray and radio emission locales is forced by the phase lag in the respective intensity peaks. The $\Delta\phi_0$ value likely receives an additional contribution from the distinct altitudes at which the polarization vectors are realized in the two wavebands. Our radio analysis suggests emission heights of order 50–100 stellar radii, whereas the polarization-limiting radius inferred for the X-ray emission likely lies at only a few tens of stellar radii (see section 'Interpreting the X-ray intensity and polarization signals'). This could be further affected by any toroidal twists to the field morphology at the respective altitudes.

The reference position angle ψ_0 inferred from the X-ray RVM fit differs from the radio value by about 15°, corresponding to a fractional difference of $\lesssim$20%. For a symmetric radio cone, the parameter ψ_0 corresponds to the projected orientation of the spin axis on the plane of the sky, modulo an additive constant associated with the dominant polarization eigenmode. For an ideal RVM in a pure dipole geometry, similar emission heights and no propagation effects, the radio and X-ray reference PAs should either be identical if the two emissions are dominated by the same polarization mode or offset by 90° if they are not. The observed difference is modest enough to suggest that the radio and X-rays are dominated by the same polarization mode. It is likely that its non-zero value could be the imprint of magnetospheric field twists. The principal signature of twists is that they rotate the PA (that is, the angle ψ_0) due to the presence of a toroidal field component. Given somewhat distinct emission heights for the two wavebands, modest variance in PA is expected[57,58].

In conclusion, the ability of the RVM to adequately describe the X-ray PA and the radio one provides an important consistency check for the VB interpretation. In the presence of strong QED birefringence, the X-ray polarization vector is expected to adiabatically track the local magnetic field direction out to the polarization-limiting radius, beyond which it becomes fixed. This naturally produces an RVM-like PA swing. The observed similarity between the X-ray and radio PA morphologies supports a scenario in which the large-scale dipolar field geometry, perhaps combined with a moderate contribution from twists, governs the polarization in both bands, with VB acting to preserve this imprint in the X-ray signal.

Interpreting the X-ray intensity and polarization signals

To model the anisotropy and polarization characteristics of X-ray emission observed for 1E 1547.0−5408, we used MAGTHOMSCATT[35,38–40], a Monte Carlo radiative transfer simulation that treats polarized scattering transport in magnetized NS atmospheres. At any point on the surface removed from the magnetic pole, the emergent radiation is highly polarized (≳80%); see ref. 35 for details on the angular distributions of local anisotropy and polarization. General relativistic light bending and redshift are included: here, we adopt a compactness value of 0.425 for an NS radius of $R_{\rm NS} = 12$ km and mass of $M_{\rm NS} \approx 1.7 M_{\odot}$. The influences of VB during magnetospheric propagation are encoded and can be switched on or off, as desired[40]. Mode conversion at the VR deeper in the atmosphere is not incorporated; we model only the softest energy band of 2–3 keV, in which the impact of VR is expected to be relatively modest.

The simulation is versatile in treating local atmospheric magnetic field directions in general relativistic dipole geometry, allowing for modelling various hotspot configurations. We explored various values of the hotspot area A and found that the conclusion about VB was not dependent on our specific choice of A. Thus, we restrict the discussion here to $A = A_{\rm BB}$ cases only, where $A_{\rm BB} = 13$ km^2 is the inferred BB emission area from the best spectral fit. Different hotspot configurations, both circular and non-circular, were considered. For circular hotspots, an area of $A_{\rm BB}$ corresponds to a half-opening angle of $\theta_{\rm cap} \approx 10^{\circ}$. For non-circular ones, we assume a wedge for simplicity. The inclination angle α ranged from 0° (aligned rotator) to 90° (orthogonal rotator) in increments of 2°. The viewing angle ζ ranged from 0° to 180°, divided into 180 bins.

Among all cases considered, the best fit to the intensity (χ^2/dof = 33.9/30), Stokes Q/I (χ^2/dof = 19.0/4) and U/I (χ^2/dof = 10.9/4) data correspond to an offset non-circular configuration, represented by a wedge with magnetic co-latitudes spanning $0 \leq \theta_m \leq 17°$ and longitudes ranging from the meridian to 120° (Fig. 3, blue region). This was obtained for $(\alpha, \zeta) = (0°, 1.5°)$ with the inclusion of magnetospheric VB; see the blue curves in Extended Data Fig. 8. We note that the dominant contribution to the χ^2 values in the Q/I and U/I fits comes from the seventh phase bin ($\phi = 0.875$), which is only slightly above the MDP_{99}. The (α, ζ) values inferred through our modelling approximately support the aligned geometry inferred by the radio intensity and the radio RVM, yet are discrepant with the X-ray RVM best-fit results. A detailed discussion of these differences is beyond the scope of this work; they may reflect a combination of non-dipolar field structure and distinct emission altitudes for the X-ray and radio signals.

The Q/I and U/I traces without VB (Extended Data Fig. 8, orange curves) show strong variations when compared with the VB-on case, a consequence of the convolution of disparate magnetic field directions on the stellar surface in conjunction with non-refractive propagation through the magnetosphere. This VB-off scenario does not provide an adequate fit to the data (Extended Data Fig. 8b, χ^2/dof = 106.8/4, and Extended Data Fig. 8c, χ^2/dof = 95.6/4). We also show in Extended Data Fig. 8d the best case among the VB-off models corresponding to a pole-centred circular hotspot (red). This too does not adequately reproduce the observed polarization data, and the corresponding geometry ($\alpha, \zeta = 2°, 20.5°$) is inconsistent with that inferred in the radio. Hence, within our adopted atmospheric modelling framework, the inclusion of magnetospheric VB provides a better description of the X-ray intensity and polarization measurements in 1E 1547.0−5408.

This inference of the likely action of VB in the magnetosphere can be made regardless of which polarization eigenmode (O mode, as in MAGTHOMSCATT, or X mode, as in refs. 2,3,59,60) emerges in dominance from the atmosphere. Using the wedge configuration (Extended Data Fig. 8), we tested this by performing simulations where the PD was presumed uniform across the surface and $\Pi_X = I_X/(I_X + I_O)$ was allowed to float as a free parameter on the interval [0, 1]. We found that the fits to the Q/I and U/I data were poor when $0.2 \lesssim \Pi_X \lesssim 0.8$, for both VB on and off, whereas fits for $\Pi_X < 0.2$ (O-mode dominance) and for $\Pi_X > 0.8$ (X-mode dominance) worked well only with VB on (and equally so), and poorly with VB off. This exploration will be detailed in a future study.

Data availability

NICER observations (ObsIDs: 8020300101, 8020300102, 8020300103 and 8020300104) and IXPE observations (04003801) are readily accessible in the HEASARC data archive: https://heasarc.gsfc.nasa.gov/W3Browse (https://doi.org/10.25504/FAIRsharing.979d22). Radio observations made by Murriyang/Parkes (ObsIDs: r070819_124237, uwl_250326_173709, uwl_250330_171906 and uwl_250331_131207; doi10.4225/08/52292AE9B2D80, doi10.25919/rzdr-pw25 and doi10.25919/v5hn-4v34) are publicly available from the CSIRO Data Access Portal (https://data.csiro.au/) following an 18-month proprietary period starting on the observation date.

Code availability

Data reduction and analysis of X-ray products were performed using publicly available software HEAsoft v.6.35.0 (https://heasarc.gsfc.nasa.gov/docs/software/lheasoft/) from the High Energy Astrophysics Science Archive Research Center (HEASARC), particularly FTOOLs v.6.35.1, SAOImage DS9 v.8.4b1 and Xspec v.12.15.0. Generation and calibration of the NICER event lists was also performed by NICERDAS v.12 of HEASoft. The simulation and analysis framework ixpeobssim v.31.1.0 was used to generate high-level IXPE data products (https://ixpeobssim.readthedocs.io/en/latest/). Moreover, the software filterbackground.py was used for the treatment of the IXPE background, found at GitHub (https://github.com/aledimarco/IXPE-background). Timing analysis was performed using tempo2 (https://github.com/mattpitkin/tempo2), PINT (https://github.com/nanograv/PINT) and CRIMP (https://github.com/georgeyounes/CRIMP/tree/main). PyXspecCorner (https://github.com/garciafederico/pyXspecCorner) and corner.py (https://corner.readthedocs.io/en/latest/) were used to generate the X-ray spectro-polarimetric and radio RVM corner plots, respectively. Additional custom code for generating figures and performing analysis is available at GitHub (https://github.com/rae-stewart/Polarimetric-Analysis-of-1E-1547.0-5408). Custom code for the MAGTHOMSCATT Monte Carlo simulation is available upon reasonable request.

42. Wilms, J., Allen, A. & McCray, R. On the absorption of X-rays in the interstellar medium. *Astrophys. J.* **542**, 914–924 (2000).
43. Verner, D. A., Ferland, G. J., Korista, K. T. & Yakovlev, D. G. Atomic data for astrophysics. II. New analytic FITS for photoionization cross sections of atoms and ions. *Astrophys. J.* **465**, 487 (1996).
44. Taverna, R. et al. The long quest for vacuum birefringence in magnetars: 1E 1547.0-5408 and the elusive smoking gun. *Astrophys. J.* **1002**, 102 (2026).
45. Everett, J. E. & Weisberg, J. M. Emission beam geometry of selected pulsars derived from average pulse polarization data. *Astrophys. J.* **553**, 341–357 (2001).
46. Ashton, G. et al. BILBY: a user-friendly Bayesian inference library for gravitational-wave astronomy. *Astrophys. J. Suppl. Ser.* **241**, 27 (2019).
47. Speagle, J. S. DYNESTY: a dynamic nested sampling package for estimating Bayesian posteriors and evidences. *Mon. Not. R. Astron. Soc.* **493**, 3132–3158 (2020).
48. Gil, J., Gronkowski, P. & Rudnicki, W. Geometry of the emission region of PSR 0950+08. *Astron. Astrophys.* **132**, 312–316 (1984).
49. Rankin, J. M. Toward an empirical theory of pulsar emission. IV. geometry of the core emission region. *Astrophys. J.* **352**, 247 (1990).
50. Johnston, S. et al. The thousand-pulsar-array programme on MeerKAT - XI. application of the rotating vector model. *Mon. Not. R. Astron. Soc.* **520**, 4801–4814 (2023).
51. Lower, M. E., Johnston, S., Shannon, R. M., Bailes, M. & Camilo, F. The dynamic magnetosphere of Swift J1818.0-1607. *Mon. Not. R. Astron. Soc.* **502**, 127–139 (2021).
52. Desvignes, G. et al. A freely precessing magnetar following an X-ray outburst. *Nat. Astron.* **8**, 617–627 (2024).
53. Foreman-Mackey, D., Hogg, D. W., Lang, D. & Goodman, J. emcee: the MCMC mammer. *Publ. Astron. Soc. Pac.* **125**, 306 (2013).
54. Doroshenko, V. et al. Determination of X-ray pulsar geometry with IXPE polarimetry. *Nat. Astron.* **6**, 1433–1443 (2022).
55. Naghizadeh-Khouei, J. & Clarke, D. On the statistical behaviour of the position angle of linear polarization. *Astron. Astrophys.* **274**, 968 (1993).
56. González-Caniulef, D., Caiazzo, I. & Heyl, J. Unbinned likelihood analysis for X-ray polarization. *Mon. Not. R. Astron. Soc.* **519**, 5902–5912 (2023).
57. Hibschman, J. A. & Arons, J. Polarization sweeps in rotation-powered pulsars. *Astrophys. J.* **546**, 382–393 (2001).
58. Tong, H., Wang, P. F., Wang, H. G. & Yan, Z. Rotating vector model for magnetars. *Mon. Not. R. Astron. Soc.* **502**, 1549–1556 (2021).
59. Lai, D. & Ho, W. C. G. Resonant conversion of photon modes due to vacuum polarization in a magnetized plasma: implications for X-ray emission from magnetars. *Astrophys. J.* **566**, 373–377 (2002).
60. van Adelsberg, M. & Lai, D. Atmosphere models of magnetized neutron stars: QED effects, radiation spectra and polarization signals. *Mon. Not. R. Astron. Soc.* **373**, 1495–1522 (2006).
61. Deller, A. T., Camilo, F., Reynolds, J. E. & Halpern, J. P. The proper motion of PSR J1550-5418 measured with VLBI: a second magnetar velocity measurement. *Astrophys. J. Lett.* **748**, L1 (2012).

Acknowledgements This work reports observations obtained with the IXPE, a joint US (NASA) and Italian (ASI) mission, led by Marshall Space Flight Center (MSFC). The research uses data products provided by the IXPE Science Operations Center (MSFC), using algorithms developed by the IXPE Collaboration (MSFC, Istituto Nazionale di Astrofisica (INAF), Istituto Nazionale di Fisica Nucleare (INFN) and ASI Space Science Data Center (SSDC)) and distributed by the High-Energy Astrophysics Science Archive Research Center (HEASARC). Murriyang, the Parkes radio telescope of CSIRO, is part of the Australia Telescope National Facility (https://ror.org/05qajvd42), which is funded by the Australian Government for operation as a National Facility managed by CSIRO. We acknowledge the Wiradjuri people as the Traditional Owners of the Observatory site. This project was supported by resources and expertise provided by CSIRO IMT Scientific Computing and made use of the Ngarrgu Tindebeek supercomputer at the OzSTAR National Facility at Swinburne University of Technology. The OzSTAR programme receives funding in part from the Astronomy National Collaborative Research Infrastructure Strategy (NCRIS) allocation provided by the Australian Government, and from the Victorian Higher Education State Investment Fund (VHESIF) provided by the Victorian Government. G.Y. acknowledges constructive discussion with A. Philippov on radio emission from magnetars and P. Ray on radio/X-ray timing analysis.

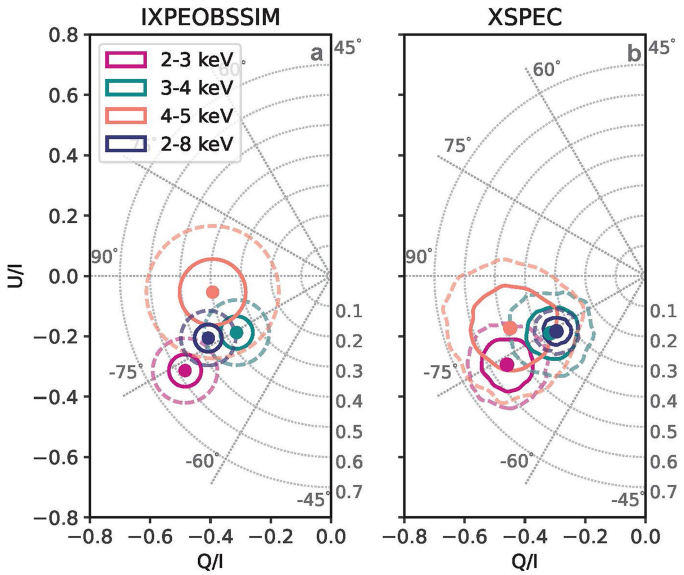
Author contributions R.E.S. performed the X-ray polarization and spectral data analysis and contributed to writing the paper. H.D.T. and M.G.B. led the simulation analysis and theoretical interpretations and contributed to writing the paper. G.Y. is the principal investigator of the IXPE observation presented in this work, which was obtained through the NASA IXPE Guest

Observer cycle 2 program. G.Y. performed the X-ray timing analysis and contributed to writing the paper. M.E.L. performed the radio observations and analysis, and contributed to writing the paper. M.N. provided support with the X-ray polarization analysis and contributed to writing the paper. F.C., J.B.C., T.E., A.K.H., W.C.G.H., C.-P.H, P.K., P.S., A.V.K. and Z.W. provided comments and contributed to the writing of the paper. F.C. is the principal investigator of the Parkes P885 project.

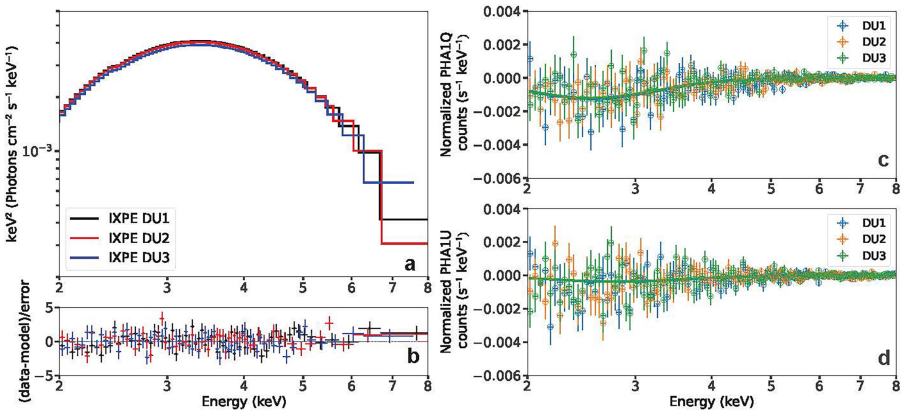
Funding The material is based on work supported by NASA under award no. 80GSFC24M0006. G.Y. acknowledges NASA support under grant nos. 80NSSC25K7257 and 80NSSC25K0283, through which R.E.S. and A.V.K. are partially supported. M.E.L. is supported by an Australian Research Council Discovery Early Career Research Award DE250100508. M.G.B. thanks NASA for generous support under grant nos. 80NSSC24K0589, 80NSSC25K7257 and 80NSSC25K0079. W.C.G.H. acknowledges support through grant no. 80NSSC23K0078 from NASA. J.B.C. acknowledges support under NASA award no. 80GSFC21M0006. F.C., A.K.H., T.E., C.P.H., P.K., M.N., P.S. and Z.W. do not declare relevant funding.

Competing interests The authors declare no competing interests.

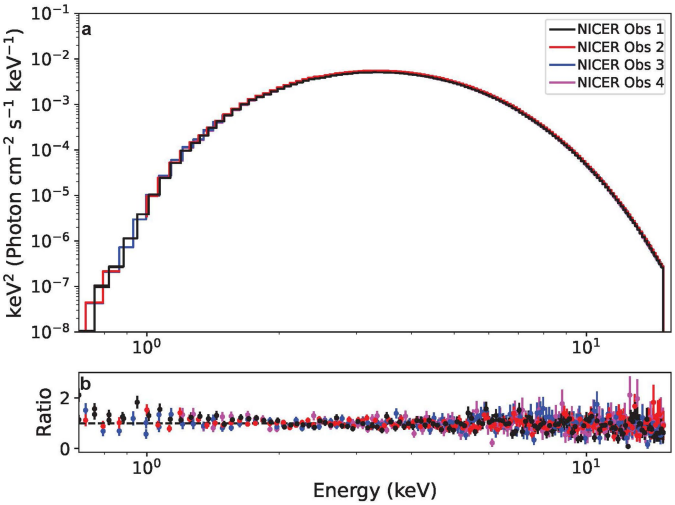
Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-026-10859-z.
Correspondence and requests for materials should be addressed to Rachael E. Stewart, Hoa Dinh Thi, George Younes, Marcus E. Lower or Matthew G. Baring.
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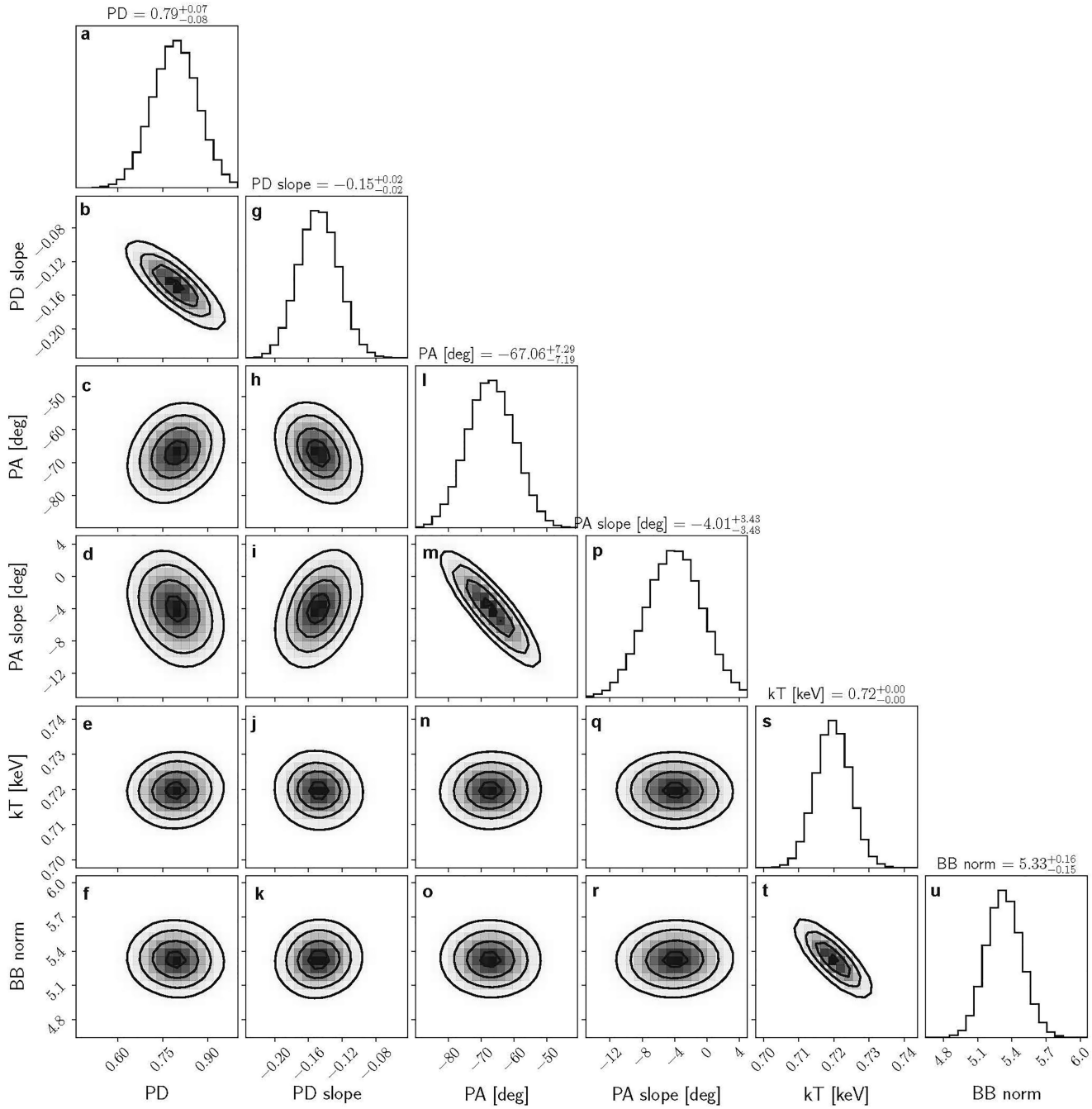
Extended Data Fig. 1 | Model-dependent and model-independent polarization characteristics comparison. A comparative study between model-independent Stokes Q and U polarization characteristics obtained through using IXPEOBSSIM (Panel a) and through applying a polconst*bbodyrad spectral model in Xspec (Panel b) at four energy bins: 2–3 keV, 3–4 keV, 4–5 keV, and 2–8 keV. The resulting PD and PA for the two methods are comparable to each other. Notably, both show a non-linear trend in the energy-dependence of the PD as the 4–5 keV band approaches the value of the 2–3 keV band at ~1σ level. Future observations with higher count statistics are needed to complement this study to more directly probe the nature of the high-energy polarization.



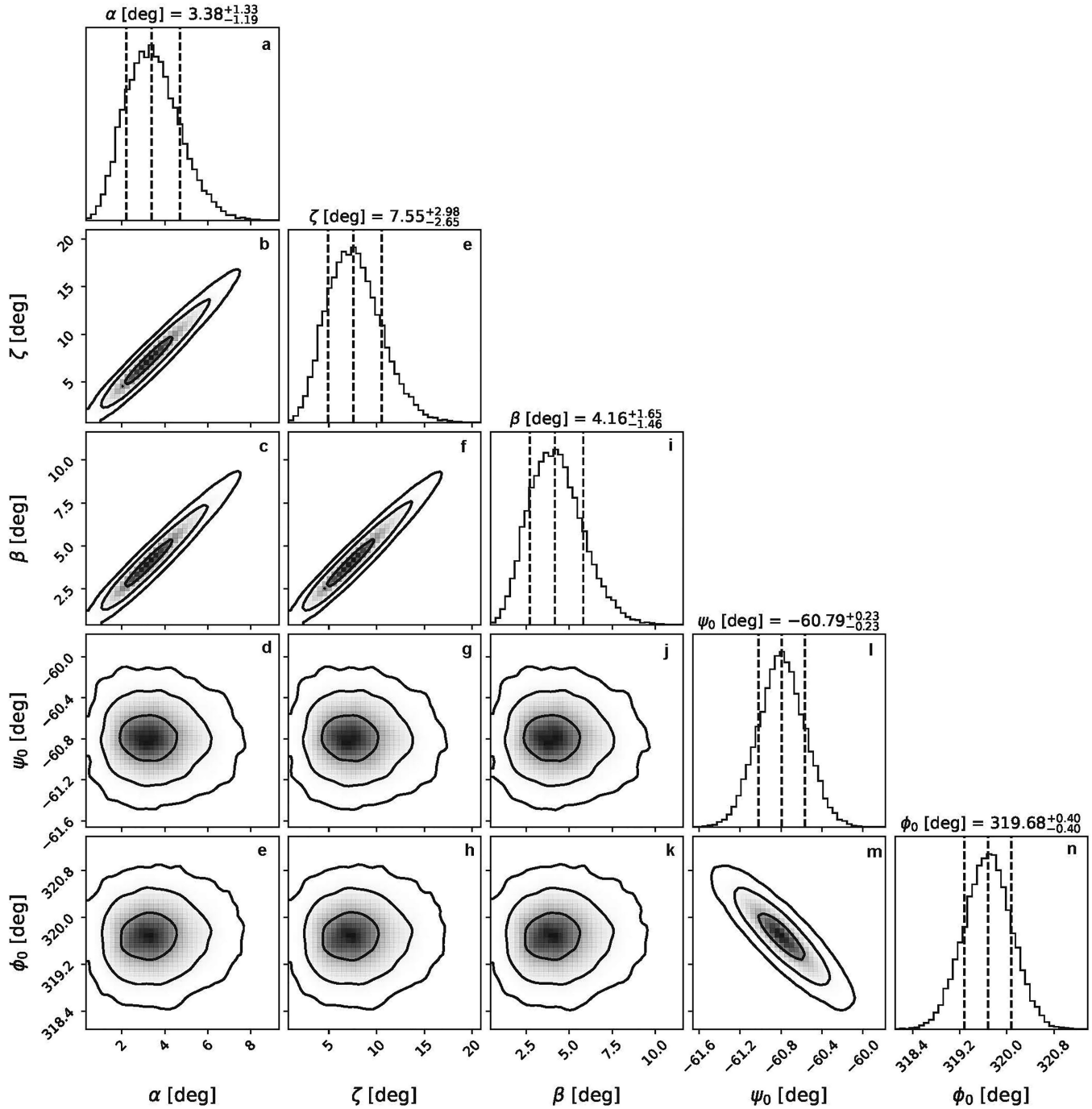
Extended Data Fig. 2 | Phase-averaged IXPE spectra. The simultaneous spectral νF_ν models of the NICER+IXPE observations. Panel a displays the best fit model, a single absorbed blackbody with a linear polarization component: constant*tbabs(pollin*bbodyrad). NB: Only the IXPE spectra are displayed here for the sake of visual clarity. Panel b shows the data divided by the folded model for the absorbed BB model. The right-hand panels show the normalized Stokes *Q*/*I* (Panel c) and *U*/*I* (Panel d) spectra in linear space for the three IXPE DUs with the solid lines showing the best fit of the linear polarization component. The quasi-thermal BB is accompanied by a strong polarization signal that decreases as a function of energy.



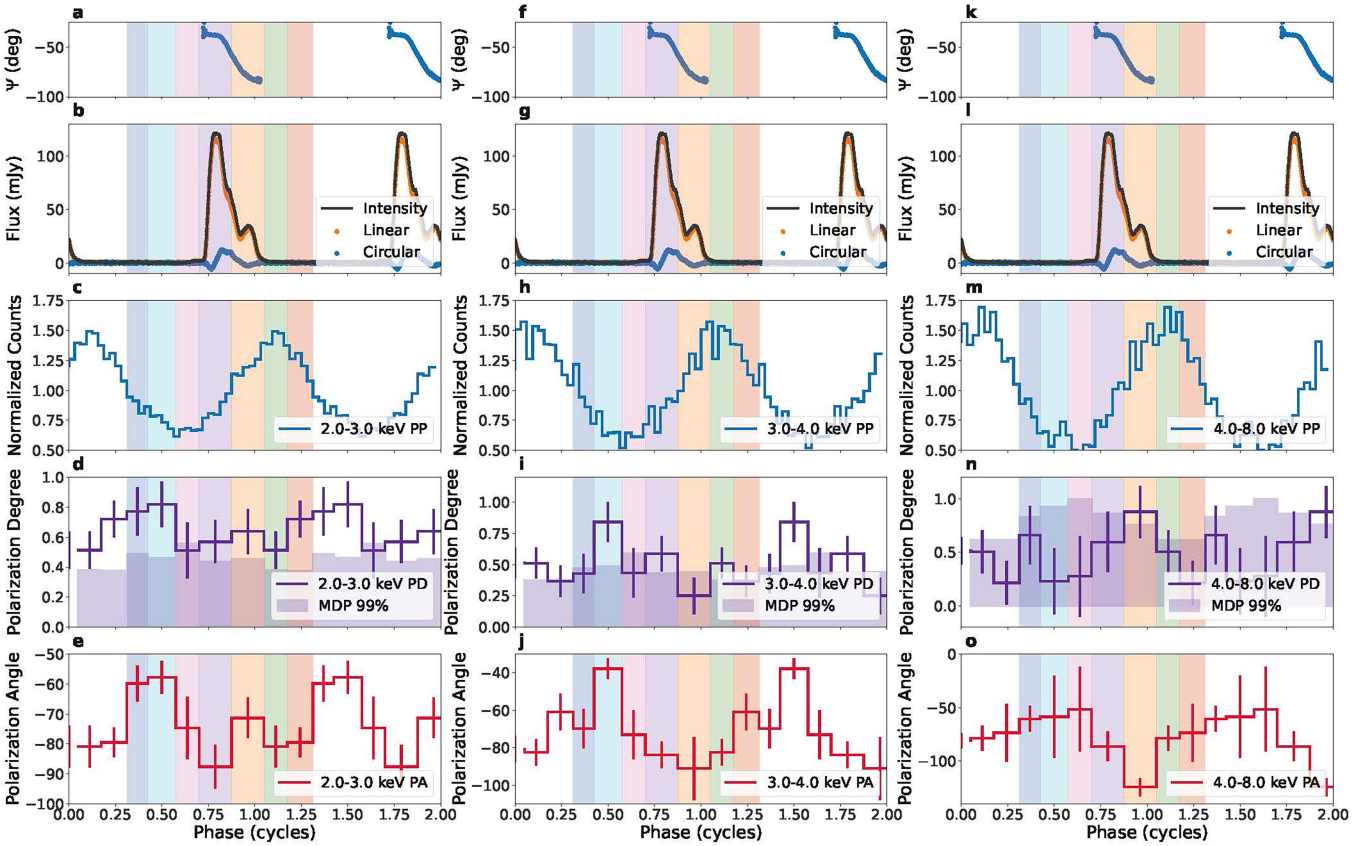
Extended Data Fig. 3 | Phase-averaged NICER spectra. Simultaneous spectral modeling of the NICER+IXPE observations in the νF_ν representation (see also Extended Data Fig. 2). The top panel shows the best-fit model to the NICER spectra, described by a single absorbed blackbody. For visual clarity, we do not display the SCORPEON background model components. The bottom panel shows the ratio of the NICER data to the folded model.



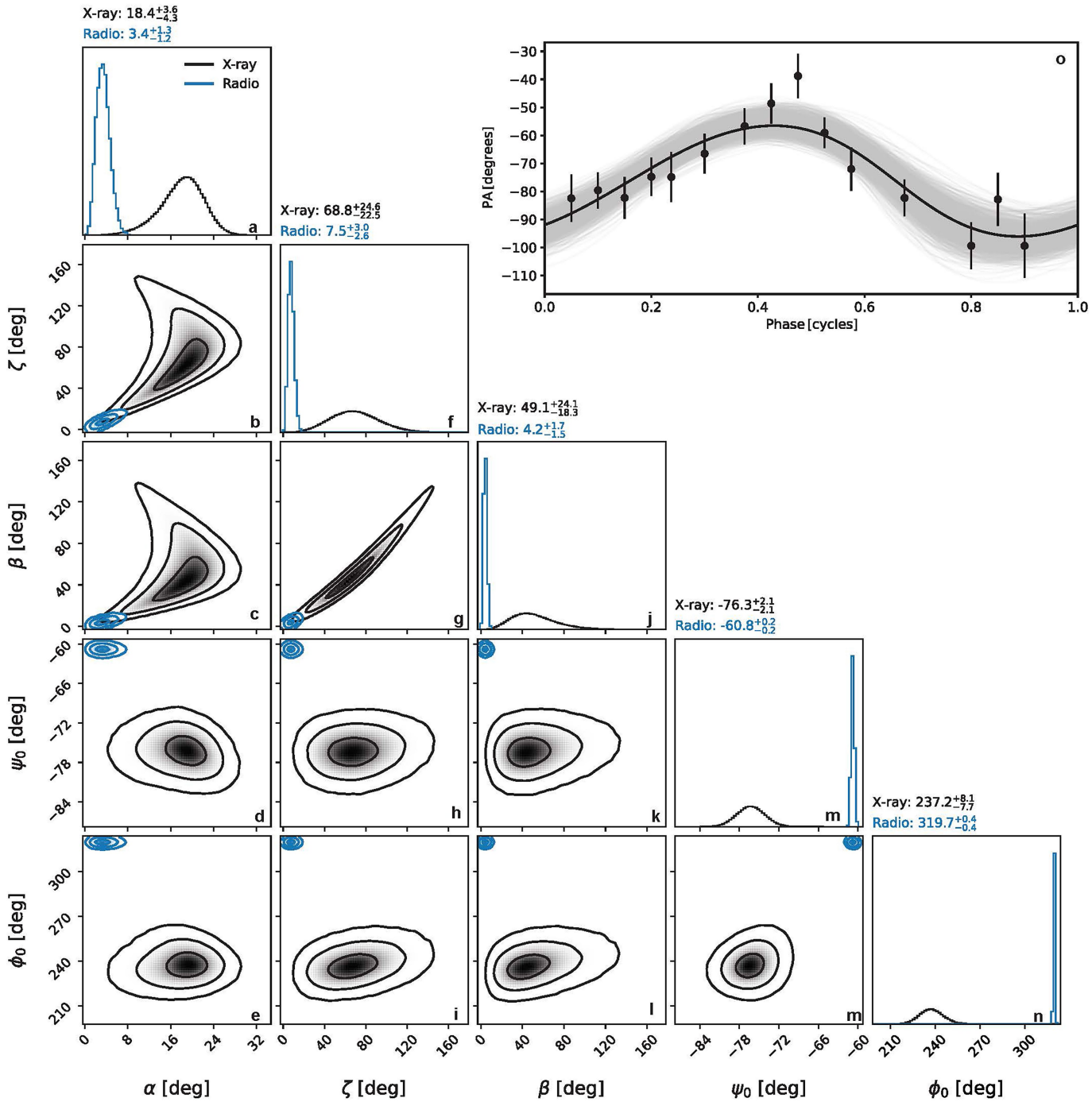
Extended Data Fig. 4 | X-ray spectro-polarimetric posterior distributions. Corner plots of the one- and two-dimensional posterior distributions of the X-ray spectro-polarimetric parameters from the best-fit model shown in Table 1 produced by an MCMC chain (Panels a-u). The contours denote the 11.8%, 39.3%, 67.5%, 86.4% credible regions (corresponding to 0.5, 1, 1.5, and 2σ).



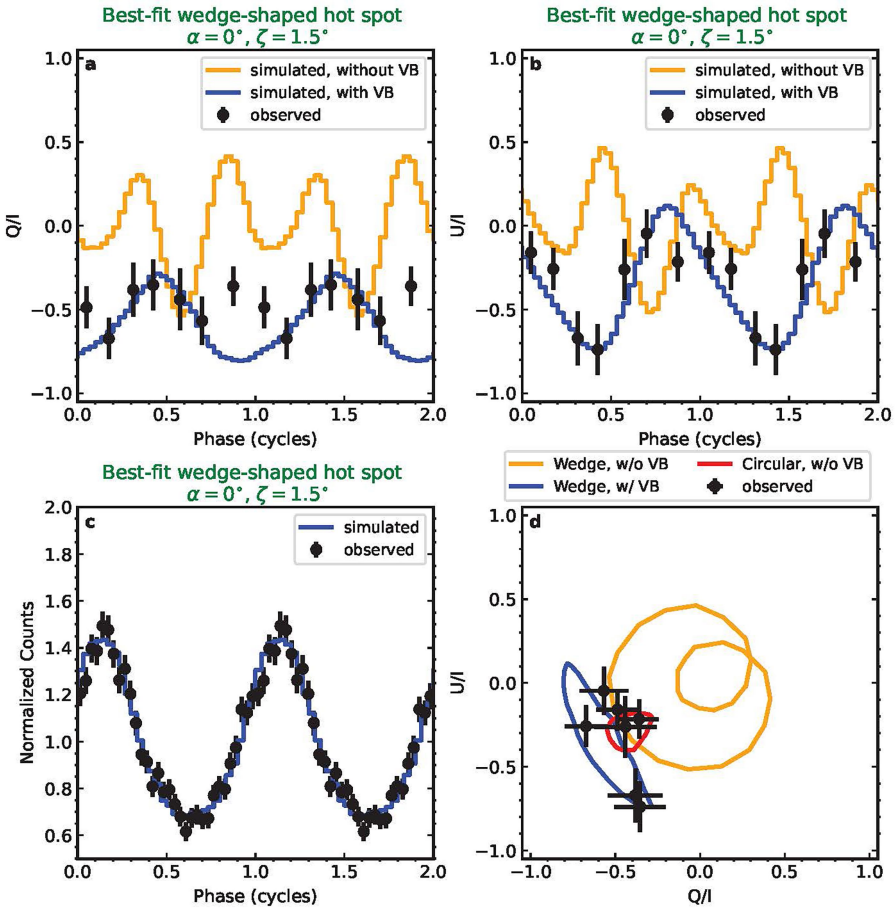
Extended Data Fig. 5 | Radio RVM posterior distributions. Corner plots of the one- and two-dimensional posterior distributions of the radio RVM fit parameters (Panels a-n). The contours denote the 39%, 87%, and 99% credible regions (corresponding to 1, 2, and 3σ).



Extended Data Fig. 6 | Phase-resolved polarization characteristics in three X-ray bands. Complementary plot to Fig. 2, with IXPE intensity, PD, and PA (shown in the lower three panels, respectively) binned according to 2–3 keV (a–e), 3–4 keV (f–j), and 4–8 keV (k–o).



Extended Data Fig. 7 | RVM fit to the phase-resolved X-ray PA. *Panels a-n*: Black contours represent the one- and two-dimensional posterior distributions for the RVM parameters $\alpha, \zeta, \beta, \psi_0$, and ϕ_0, obtained from a Bayesian fit to the 2-4 keV phase-resolved polarization angle profile using the[55] wrapped-angle likelihood. Contours denote the 39%, 87%, and 99% credible regions (corresponding to 1, 2, and 3σ). The blue contours are those derived from the radio RVM fit overlaid for ease of comparison. *Panel o*: Phase-resolved X-ray PA measurements (black points) overlaid with the maximum-posterior RVM model (solid black line). Light gray curves show random posterior draws, illustrating the range of model realizations consistent with the data.



Extended Data Fig. 8 | Stokes Q/I, U/I, and I modeling. Simulated Stokes *Q*/*I* (panel a), *U*/*I* (panel b), and intensity (panel c) pulse profiles from MAGTHOMSCATT for a single hotspot wedge offset from the magnetic pole as a function of rotational phase (solid lines). The black dots represent the Stokes *Q*/*I* (panel a), *U*/*I* (panel b), and intensity (panel c) data extracted from IXPE in the 2–3 keV energy range. The case with the best statistical fit (lowest combined total χ^2; see Methods) incorporates vacuum birefringence, and corresponds to a magnetic colatitude of $\theta_m = [0°, 17°]$ and longitude $\phi_m = [0°, 120°]$ (zero longitude contains the rotation and magnetic axes); it is displayed in blue. The orange solid curves on the center and right panel show the corresponding polarization profiles wherein VB is turned off - these fits are statistically worse than those with VB on (see Methods). Panel (d) displays the comparison in the Stokes Q-U space between the observed data (black dots) and the simulated results obtained with the best-fit wedge-shaped hotspot with (blue line) and without (orange line) including magnetospheric VB, plotted for one rotational cycle. The red solid line represents the best result among the VB-off cases (see Methods), corresponding to a pole-centered circular hotspot with a magnetic inclination of $\alpha = 2°$ and a viewing angle of $\zeta = 20.5°$.

Extended Data Table 1 | Timing and Pulsar Parameters

Pulsar parameters	
R. A. (J2000) [a]	$15^h\ 50^m\ 54.124$
Decl. (J2000) [a]	-54°18' 24.11
ν(Hz)	0.477856(3)
$\dot{\nu}(s^{-2})$	-4.4×10^{-12}
Timing epoch (MJD TDB)	58350
Position epoch (MJD TDB)	54795
DM (pc cm^{-3})	697(1)
Time span (MJD)	60760–60770
Solar system ephemeris	DE405
TZRMJD	60764.7495
TZRFRQ	3275.71
TZRSITE	pks

Timing and pulsar parameters of the magnetar 1E 1547.0 – 5408 from joint *Murriyang* + NICER observations. Values in parentheses are the 1σ uncertainties on the last digit. [a] Fixed to values from[61].

Extended Data Table 2 | Observations

Telescope	Obs. ID	Start (UTC)	Stop (UTC)	Livetime (ks)
IXPE	04003801	2025-03-26T03:01:52	2025-04-05T00:26:53	499.5
NICER	8020300101	2025-03-26T01:09:00	2025-03-26T15:13:00	1.057
NICER	8020300102	2025-03-27T01:51:01	2025-03-27T20:40:58	1.42
NICER	8020300103	2025-03-28T14:56:37	2025-03-28T22:47:32	1.051
NICER	8020300104	2025-03-31T06:33:41	2025-03-31T08:16:16	0.996
Murriyang	r070819_124237	2007-08-19T12:42:37	2007-08-19T12:17:06	0.869
Murriyang	uwl_250326_173709	2025-03-26T17:37:09	2025-03-26T18:29:50	3.171
Murriyang	uwl_250330_171906	2025-03-30T17:19:06	2025-03-30T17:59:36	2.436
Murriyang	uwl_250331_131207	2025-03-31T13:12:07	2025-03-31T13:59:53	2.873

List of Parkes/Murriyang, IXPE, and NICER observation IDs, observation start and end times, and total livetimes used in this analysis.